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Session Objectives

At the end of this session, you will be able to

1. Discuss the etiopathogenesis and morphology of **congenital anomalies** of the ureter and the bladder
2. Describe the etiopathogenesis and morphology of **infectious, reactive and inflammatory lesions** of the ureter, bladder and urethra
3. To become familiar with the etiopathogenesis and morphology of **neoplastic lesions** of the ureter, bladder and urethra.

Please note that unless mentioned otherwise, all the images and information included in this presentation are from Robbins and Cotran, Pathologic Basis of Disease, 10th edition

Ureters: Congenital Anomalies

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VUR Reflux
Double and Bifid Ureters
Ureteropelvic junction obstruction

These are seen in 2% to 3% of all autopsies. Most have little clinical significance;

Vesicoureteral reflux (VUR)

It represents retrograde urine flow from the urinary bladder to the upper urinary tract.
Induces severe nephropathy. Early treatment is very important.
Voiding cystourethrogram (VCU) is the gold standard method of diagnosis

Pathogenesis: Steps

The ureter runs downwards and drains urine from the kidney into the urinary bladder.
Intramural/intravesical ureter: an oblique tunnel within the bladder.
Its orifice typically remains closed due to its obliquity.
The ureteric peristalsis increases the ureteric pressure.
Subsequently, the orifice opens to allow the passage of urine into the bladder.
Then the ureteric pressure drops while the intravesical pressure rises again and closes the intramural ureteral part.

The distal ureteric orifice remains open if the ratio of intramural tunnel length to the ureteral diameter is less than 5.
Also seen in immature bladder discoordination/ bladder dysfunction.

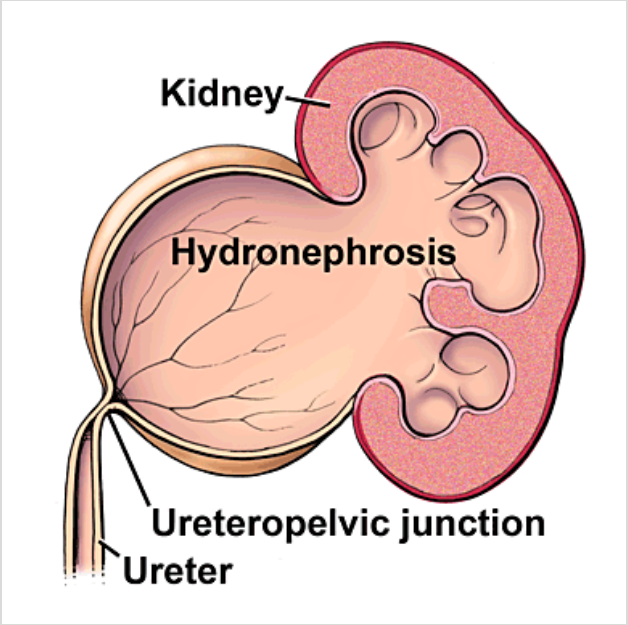
Ureteropelvic junction (UPJ) obstruction

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It is the most common cause of hydronephrosis in infants and children which preferentially affects males and is bilateral in 20% of cases.
There is agenesis of the contralateral kidney in a minority of cases.

In adults: more common in women and is most often unilateral.

There is abnormal organization of smooth muscle bundles or excess stromal deposition of collagen between smooth muscle bundles at the UPJ obstruction.
Other cases are determined by extrinsic compression of the UPJ by abnormal renal vessels.



Ureter: Inflammatory conditions

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Ureteritis follicularis: Long standing cases, accumulation of lymphocytes in subepithelial region causes slight elevation of the mucosa

Ureteritis cystica: cysts on the mucosa.
It is asymptomatic.
It results from long-standing chronic inflammation.
Nests of transitional epithelium grow downward into the lamina propria and a central cystic space forms



Ureteritis cystica: opened ureter with multiple small glistening "bumps" on the surface.

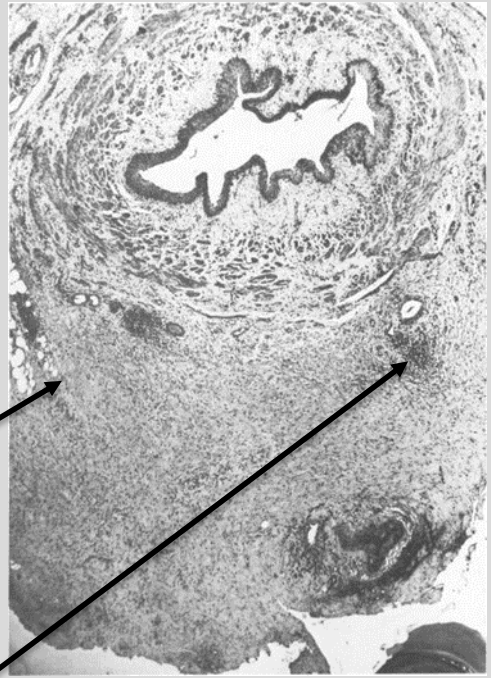
Sclerosing Retroperitoneal Fibrosis

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It is a fibrotic inflammatory process that encases retroperitoneal structures and causes hydronephrosis.
-middle to late age and is more common in males than females.

It is associated with drug exposures (ergot derivatives, β-adrenergic blockers), inflammatory conditions (vasculitis, diverticulitis, Crohn disease), or malignancies (lymphomas, urinary tract carcinomas).
Most, however, have no obvious cause and are considered primary or idiopathic SRF.

Microscopic examination: fibrous tissue containing a prominent infiltrate of lymphocytes, often with germinal centers, plasma cells and eosinophils.



PMC full text:
Ann Surg. 1972 Aug; 176(2): 199–204.
doi: 10.1097/00000658-197208000-00013

Ureter: Malignant tumors

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Secondary malignant tumors of the ureter are metastases of renal and bladder primaries.

Primary malignant tumors of the ureter are similar with those arising in the renal pelvis, calyces, and bladder.
The majority are urothelial carcinomas.
They occur most frequently during the sixth and seventh decades of life and cause obstruction of the ureteral lumen.
They commonly occur concurrently with urothelial carcinomas of bladder or renal pelvis.



Papillary transitional cell carcinoma extensively involving the ureter.
(Courtesy Dr. Cristina Magi-Galluzzi, The Johns Hopkins Hospital, Baltimore, MD.)

Urinary Bladder: Congenital Anomalies

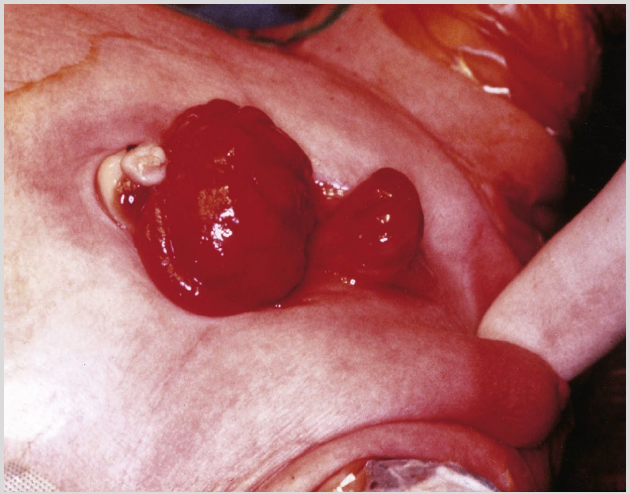
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Diverticula: 1 cm to 10 cm pouch-like invaginations of the bladder wall. Represent a risk for infection and bladder calculi.

- **Congenital diverticula** are caused by a focal failure of development of the normal musculature or to a urinary tract obstruction during fetal development.
- **Acquired diverticula** are associated with prostatic hyperplasia, producing urinary outflow obstruction. The resulting increase in intravesical pressure causes outpouching of the bladder wall and the formation of diverticula.

Exstrophy of the bladder

The bladder communicates directly with the abdominal surface. The exposed bladder mucosa may undergo glandular metaplasia and is subject to chronic infection that often spreads to the upper urinary tract



Exstrophy of the bladder in a newborn boy. The tied umbilical cord is seen above the hyperemic mucosa of the everted bladder. Below is an incompletely formed penis.
(Courtesy Dr. John Gearhart, The Johns Hopkins Hospital, Baltimore, MD.)

Urinary Bladder: Inflammation
Acute and Chronic Cystitis

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Bacterial pyelonephritis is frequently preceded by infection of the urinary bladder.

Predisposing factors:
bladder calculi, urinary obstruction, diabetes mellitus, immune deficiency, instrumentation

Etiology
Coliforms: Escherichia coli, Proteus, Klebsiella, Enterobacter.
Viruses (adenovirus), Mycoplasma.
Schistosomiasis (Schistosoma haematobium): cystitis in certain African and Middle Eastern countries.
Tuberculous cystitis after renal tuberculosis.

Women are more likely to develop cystitis as a result of their shorter urethras

Obstruction of the bladder outlet

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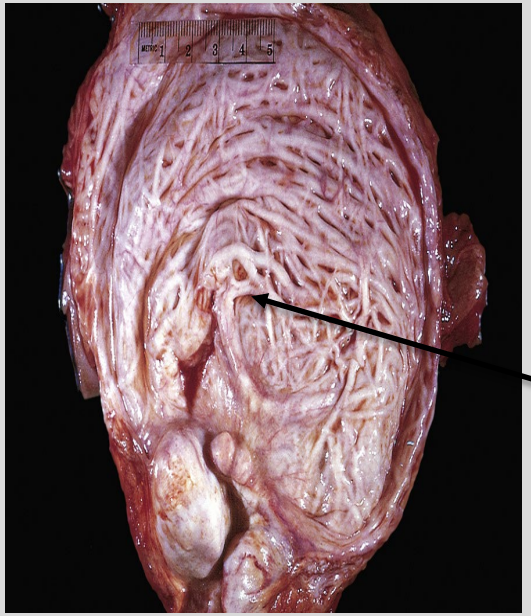
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It has a major clinical importance because of its eventual effect on the kidney.

Common in men; caused by BPH
Women: caused by cystocele of the bladder.

Other causes:

- congenital urethral strictures.
- inflammatory urethral strictures.
- bladder tumors, either benign or malignant.
- mechanical obstructions caused by foreign bodies and calculi.
- injury of nerves controlling bladder contraction (neurogenic bladder).



Morphology

Early: thickening of the bladder wall due to smooth muscle hypertrophy.

Long standing: the individual muscle bundles greatly enlarge and produce trabeculation of the bladder wall

Urothelial Neoplasms Risk Factors

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Cigarette smoking
- the most important etiologic factor, increasing the risk 3-7X

Industrial exposure to aryl amines
-2-naphthylamine and related compounds
-Cancer appears 15 to 40 years after the first exposure.

Schistosoma haematobium infections
-endemic areas (Egypt, Sudan); the organism determines a chronic inflammatory response promoting a progressive mucosal dysplasia, and neoplasia.
-70% of these cancers are squamous, the remainder being urothelial or adenocarcinoma

Long-term use of analgesics
Heavy long-term exposure to cyclophosphamide
Irradiation

Urothelial Neoplasms Pathogenesis

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Non–invasive papillary cancers

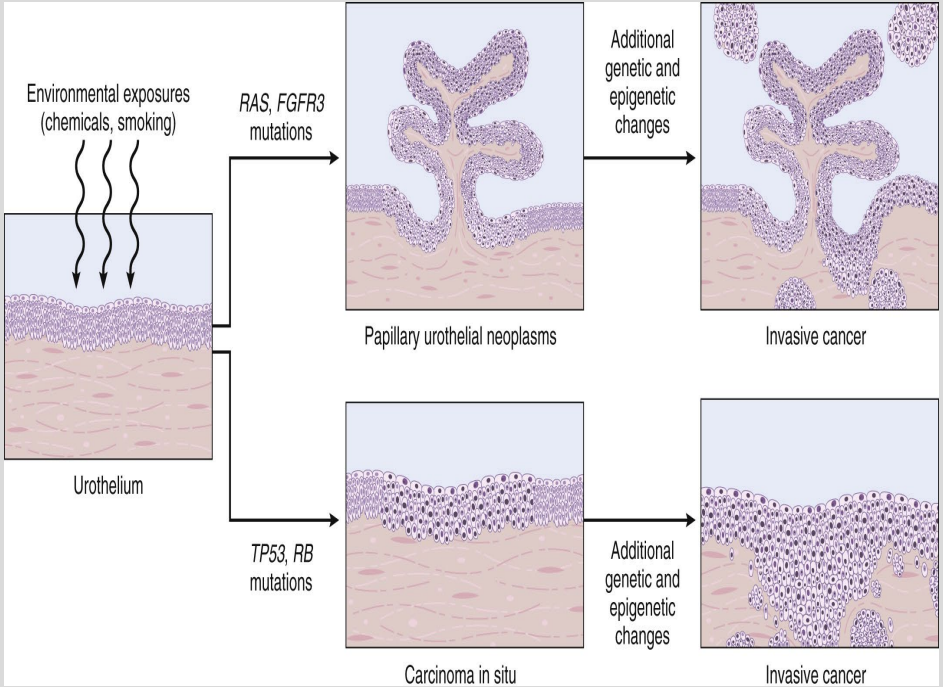
-Present with gain-of-function alterations that increase signaling through growth factor receptor pathway:

a. amplifications of the FGFR3 tyrosine kinase receptor gene and
b. activating mutations in the genes encoding RAS and PI 3-kinase.

-These tumors frequently recur but progress to muscle-invasive bladder cancer in only about 20% of cases.

The majority of muscle-invasive bladder cancers develop by progression from "flat" CIS.

- These lesions have mutations of p53 and RB
- These mutations occur early in the development of CIS and later in the progression of papillary cancers.
- high grade as they is multifocal



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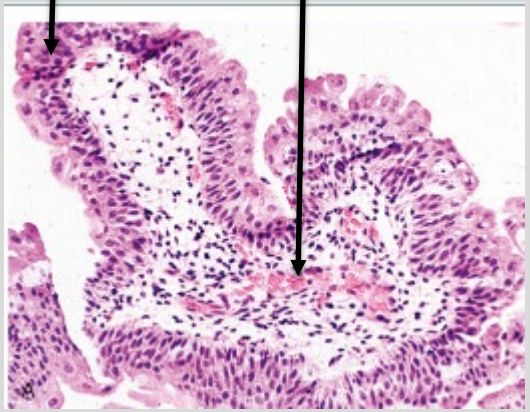
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Papillomas

1% or less of bladder tumors, often seen in younger patients. .

The individual finger-like papillae have a central core of loose fibrovascular tissue covered by epithelium that is histologically identical to normal urothelium.

Recurrences and progression are rare



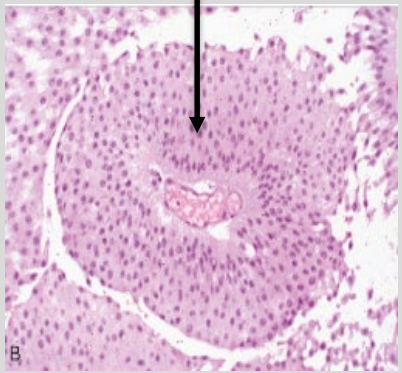
Papilloma consisting of small papillary fronds lined by normal-appearing urothelium.

Papillary urothelial neoplasms of low malignant potential (PUNLMP)

These are vey similar with papillomas, differing only in having thicker urothelium with greater density of cells.

They are larger than papillomas .

Progression to tumors of higher grade may occur but is rare.



Papillary urothelial neoplasms of low malignant potential (PUNLMP) : thicker urothelium with greater density of cells.

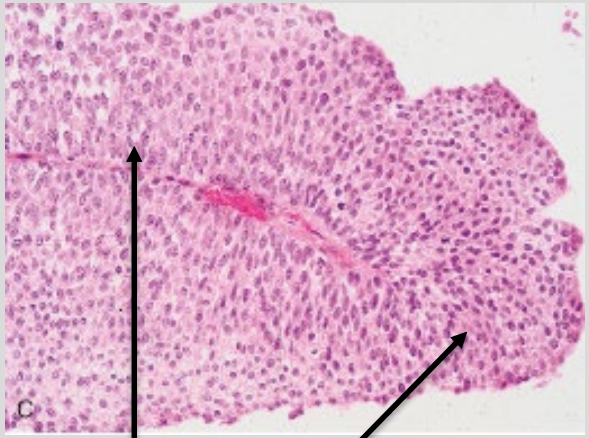
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Low-grade papillary urothelial carcinomas

They have an orderly architectural appearance showing low-grade cytologic atypia.

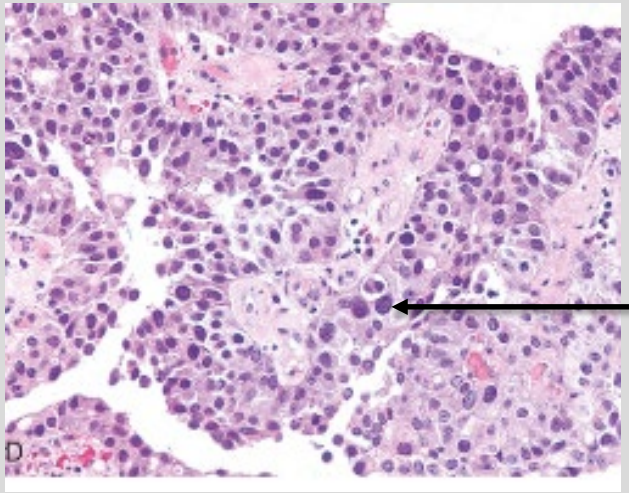
These low-grade cancers may recur but invade invade rarely.



Low-grade papillary urothelial carcinoma with an overall orderly appearance, with scattered hyperchromatic nuclei

High-grade papillary urothelial carcinomas

There is disarray and loss of polarity, and they have a greater incidence of progression to muscle-invasive bladder cancer. Also, they have a significant potential for metastasis to regional lymph nodes and systemic spread to liver and lung.



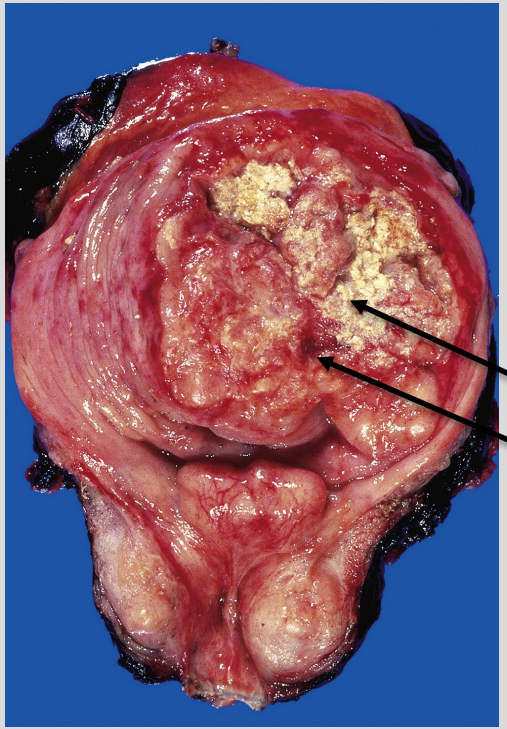
High-grade papillary urothelial carcinoma with marked cytologic atypia: hyperchromasia and prominent nucleoli.

Invasive urothelial carcinoma

The most important prognostic factor is extent of spread-*STAGE*:
it is based primarily on depth of invasion in the bladder wall, at the time of initial diagnosis.

Stage determines treatment modality: invasion of the muscularis propria layer is an indication for radical cystectomy or radiation therapy or adjuvant chemotherapy.

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Opened bladder showing a high-grade invasive transitional cell carcinoma at an advanced stage:
This is an aggressive multinodular neoplasm has fungated into the bladder lumen and spread over a wide area.
Necrosis
Ulceration

Pathology of the Urethra: Inflammation

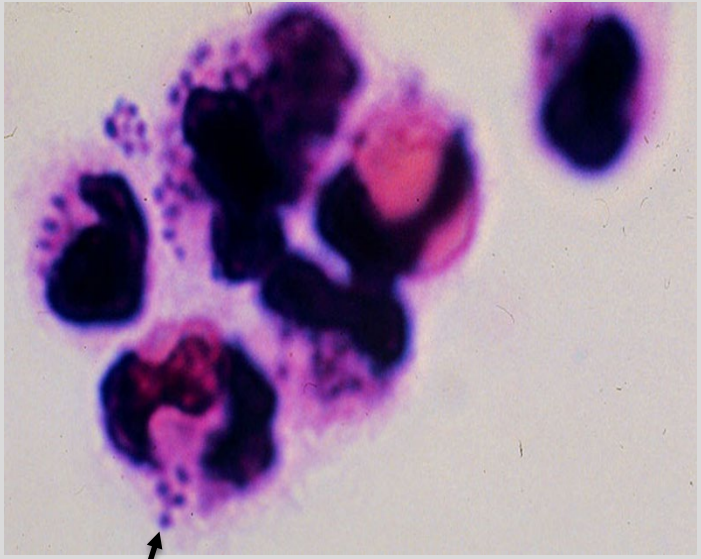
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Gonococcal urethritis is one of the earliest manifestations of this venereal infection.
Nongonococcal urethritis is common and can be caused by several different organisms.

- Various strains of Chlamydia (Chlamydia trachomatis) : 25% to 60% of nongonococcal urethritis in men and about 20% in women.

Urethritis is often accompanied by cystitis in women and by prostatitis in men.

Non-infectious urethritis: In Reactive Arthritis aka Reiter syndrome:
-clinical triad of arthritis, conjunctivitis, and urethritis



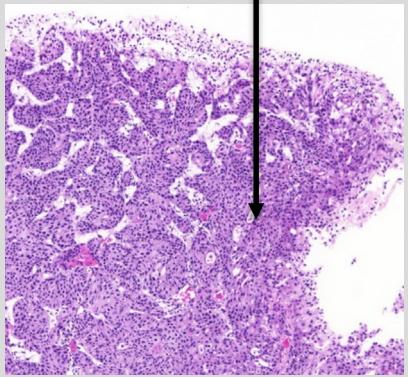
Urethritis discharge: PMNs containing diplococci corresponding to **N. gonorrhoeae.**

Tumor-Like lesions and tumors

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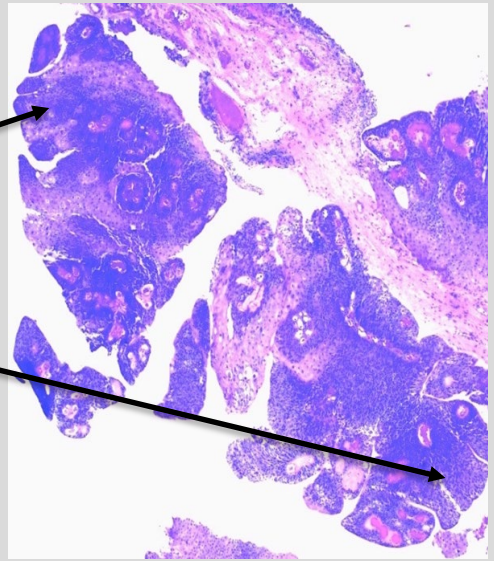
Urethral caruncle is an inflammatory lesion:
-small, red, painful mass about the external urethral meatus, typically in older females.
-inflamed granulation tissue covered by an intact but friable mucosa which ulcerates and bleeds with the slightest trauma.
-surgery provides cure.

Benign epithelial tumors of the urethra include urothelial papillomas, inverted urothelial papillomas, and condylomas.



Inverted urothelial papilloma: Urothelium invaginates into the lamina propria

Condyloma: Polypoid lesion - of hyperplastic papillary fronds of squamous epithelium with koilocytosis-viral damage (not seen at the low power magnification).
It is caused by HPV (both low risk HPV and high-risk HPV).



Primary carcinoma of the urethra

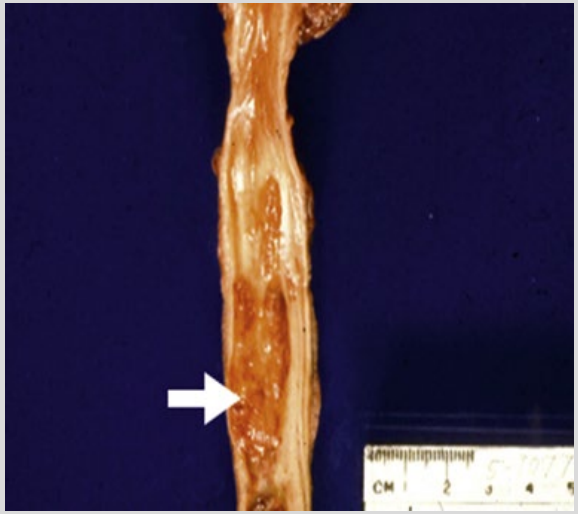
1% or urinary tract malignancies: **Urothelial and Squamous**

Proximal urethra: urothelial differentiation and are analogous to those occurring within the bladder

Distal urethra: squamous cell carcinomas and HPV-related.

Overall, urethral carcinoma is aggressive.
It determines metastasis in: lung, liver, bone.

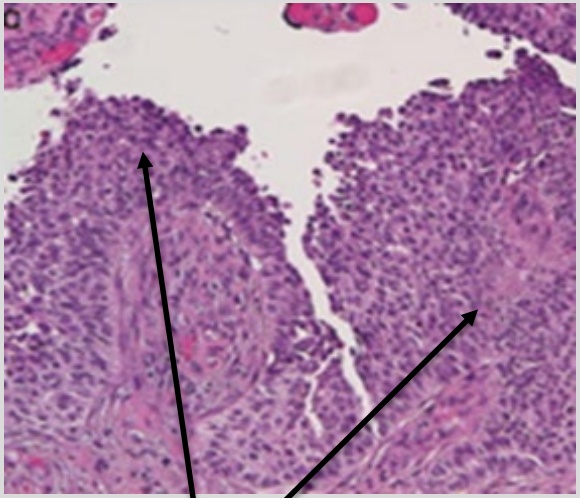
Adenocarcinomas are infrequent in the urethra and generally occur in women



Carcinoma of urethra with a fungating growth pattern (arrow).

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SCC of the urethra: hyperchromatic malignant cells below basement membrane, with active nuclei and visible nucleoli (at high power), atypical mitosis.